DG INTERNS HUB
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IP, MAC, DNS, DHCP and Common Ports

1. Core Networking Terms

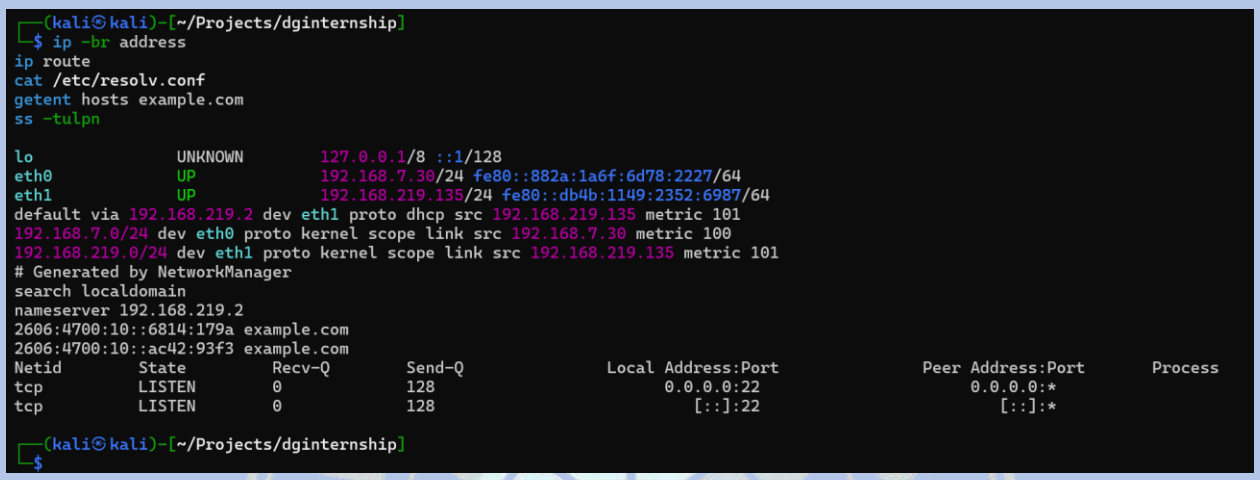
- **IP address:** A logical address used to identify an interface and route traffic across IP networks. IPv4 uses 32 bits, while IPv6 uses 128 bits.
- **MAC address:** A link-layer address used in local network communication. Hardware addresses can sometimes be changed or randomized.
- **DNS:** The Domain Name System translates domain names into records such as IP addresses and also supports other record types.
- **DHCP:** Dynamic Host Configuration Protocol automatically provides network settings such as an IP address, subnet mask, default gateway and DNS servers.
- **Port:** A transport-layer identifier used by TCP or UDP to direct traffic to a service or application endpoint.

2. Common Ports and Typical Risks

Port	Common service	Typical risk if exposed or misconfigured
21/TCP	FTP control	Plaintext credentials or insecure configuration
22/TCP	SSH	Password guessing, stolen keys or weak authentication
23/TCP	Telnet	Credentials and traffic may be transmitted without encryption
25/TCP	SMTP	Open relay abuse or mail-server misconfiguration
53/TCP/UDP	DNS	Misconfiguration, cache poisoning risks or amplification abuse
80/TCP	HTTP	Unencrypted web traffic; vulnerable web applications
110/TCP	POP3	Unencrypted credentials if TLS is not used
143/TCP	IMAP	Unencrypted credentials if TLS is not used
443/TCP	HTTPS	Application vulnerabilities, weak configuration or compromised certificates
445/TCP	SMB	Unauthorized file sharing, weak access controls or unpatched services

3306/TCP	MySQL	Unauthorized database access or excessive network exposure
3389/TCP	RDP	Credential attacks or exploitation of vulnerable remote-access services

3. Example Networking Commands



4. DNS Resolution Flow

1. The user enters a domain name in a browser.
2. The operating system checks relevant local caches and configured name-resolution sources.
3. A DNS resolver obtains the required record if it is not already cached.
4. The resulting IP address is returned to the client.
5. The client attempts to connect to the destination service.

5. DHCP Process

The common DHCPv4 process is DORA:

- Discover
- Offer
- Request
- Acknowledge

DHCP helps devices obtain suitable network configuration without manually assigning every setting.

6. Conclusion

IP addresses, MAC addresses, DNS, DHCP and ports are foundational networking concepts. Understanding them enables a SOC analyst to interpret traffic, investigate suspicious connections and identify unsafe service exposure.

